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Effects of mobile apps for nursing students: learning motivation, social interaction and study performance

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ABSTRACT

This study examined the effects of mobile apps on the learning motivation, social interaction and study performance of nursing students. A total of 20 students participated in focus group interviews to collect feedback on their use of mobile apps for learning and communicative activities. Two consecutive cohorts of students in a nursing programme, each with about 200 students, were recruited for assessing their learning motivation through a survey and study performance by two tests. The results showed that the students actively used the mobile apps for studying supplementary materials as well as participating in in-class activities and clinical assessments. The students had a relatively high level of motivation for performing and learned well, but relatively low perceived satisfaction and self-efficacy with mobile learning. They showed better study performance after practising mobile learning. The results also suggest that students' perceived intrinsic value with mobile learning has to be enhanced for improving study performance.

KEYWORDS

Mobile learning;
nursing education;
learning motivation;
social interaction; study
performance

Introduction

The use of mobile technology is becoming an integral part of education. According to a McGraw-Hill Education's report in 2015 (McGraw-Hill Education, 2015), almost two-thirds of US college students used mobile devices for learning purposes – a 20% increase since 2014 and a 69% increase since 2013; and the advantage mentioned most frequently by these students (64%) was mobility. Additional benefits of mobile learning have been widely reported, such as enhancing interaction and collaborative learning (Project Tomorrow, 2010), and student engagement and access for learning (Aderinoye, Ojokheta, & Olojede, 2007). Together with the availability of network infrastructure, these advantages serve as the key reasons for educational institutions to practise mobile learning (UNESCO, 2013).

However, the reports on such benefits are less clear about in what ways, and to what extent, mobile learning, compared with conventional classroom learning, can enhance learning motivation, social interaction and study performance. Motivation has long been

identified as a necessary precondition for students to engage in learning activities and a factor influencing learning effectiveness (Malone & Lepper, 1987). Vogel, Kennedy, and Kwok (2009) also emphasise the significance of students' motivation in engaging in and sustaining the use of mobile devices for learning. However, despite findings on the overall changes in learning motivation after practising mobile learning being widely reported (e.g. Ciampa, 2014; Gregory & Crompton, 2016; Koole, McQuilkin, & Ally, 2010), the amount of attention given to the potential relationship between specific aspects of motivation and mobile learning has been minimal (Chan, 2002; Khaddage et al., 2015). For social interaction, instant messaging mobile apps have become prevalent among students. However, their effects have been examined less in specific learning contexts, such as nursing education which involve substantial face-to-face and distance communication among students, instructors and clinical mentors. Also, how student performance is influenced by motivational factors in mobile learning practices has yet to be established.

This paper presents the results of a study on the effects of mobile learning on students' learning motivation, social interaction and study performance. It focuses on the use of mobile apps in nursing education, which features a diversity of learning environments from classrooms to laboratories and wards, and a substantial part of clinical practicums. The results of this study contribute to the effective design and implementation of mobile learning for motivating students to engage in learning activities.

Related studies

Learning motivation

Prior research on mobile learning has shown that its effect on students' learning motivation is a critical issue. For example, Wang, Shen, Novak, and Pan (2008) suggested that the implementation of mobile learning in a Chinese college had changed passive students to engaged learners by promoting real-time interaction and providing information for pedagogical adjustment. Kinash, Brand, and Mathew (2012) introduced mobile learning to an undergraduate course and found that the use of mobile devices had a positive influence on students' learning motivation. Huang, Yang, Chiang, and Su (2016) compared the effectiveness of mobile learning tools with conventional learning tools, and reported that the convenient functions supported by mobile devices could create more interesting learning environments and promote students' motivation. Also, Rau, Gao, and Wu (2008) studied how mobile learning tools, especially communication technology, affected students' learning motivation, and found that a public communication environment (e.g. public forum) and the private dialogue between student and instructor (e.g. instant messaging and email) could enhance the extrinsic motivation of students.

Although students' attitudes towards the use of mobile devices for learning have been found to be positive in general, this does not guarantee the sustainable practice of mobile learning (Garrett & Jackson, 2006; Kenny, Park, van Neste-Kenny, & Burton, 2010). For example, Vogel et al. (2009), in an eight-week study of students' use of mobile devices for learning, found that the usage rate could be maintained at a high level only at the beginning of the study period. Li and Wang (2012) pointed out that 'all such efforts and research are futile or at least substantially discounted, if students are disengaged' (p. 1).

Social interaction

The communication and sharing of data, files and messages among students and teachers through the use of mobile technologies have been shown to foster student–student and student–teacher interaction. Kukulska-Hulme and Traxler (2007) pointed out that learners can be virtually connected by mobile technology and interact with other learners regardless of time and location. Similarly, Mann and Reimann (2007) suggested that the significance of mobile learning rests in its feature of enhancing interaction, apart from enabling students and teachers to act inside as well as outside the classroom and to have access to various types of resources. Tu, McIsaac, Sujo-Montes, and Armfield (2012) proposed three aspects, namely social, communication and network, for examining mobile social interaction so as to comprehend its unique features. The social aspect touches on the mobile interaction which engages learners. The communication aspect refers to multiple communication channels in which learners engage in the mobile interaction. The network aspect relates to the capability of networking technologies to link up content, people and technologies so that a context-rich learning environment can be created.

Although it has been argued that such kinds of interaction brought by mobile learning may not only have positive impacts but also negative ones (Rau et al., 2008), Järvelä, Näykki, Laru, and Luokkanen (2007) argued that engaging students in interaction and sharing is desirable and mobile learning can help to achieve such a goal. The relevance of social interaction in students' learning is reflected in its association with thorough learning (Kreijns, Kirschner, & Jochems, 2003), implying that students' learning outcomes would be better with thorough social interaction with others in class. It is therefore of interest to investigate the ways in which the implementation of mobile learning spurs interaction among students and teachers, and the effects of such interaction on students' study performance.

Study performance

There is a lack of consensus on the effect of mobile learning on learners' study performance. For instance, Vogel et al. (2009) found that the range of mobile apps developed to facilitate students' learning at a university in Hong Kong significantly improved the examination performance of those who had actively used mobile devices for learning, despite the problem of students' diminishing engagement throughout the pilot project. McConatha, Paul, and Lynch (2008) conducted an empirical study on the practice of mobile learning in a sociology course in a mid-sized university in the United States, and suggested that students who adopted mobile learning to review and study for examinations had performed better than those who did not. Amry (2014) tested the implementation of WhatsApp mobile learning activities at university, and showed the positive effect of these activities on students' performance.

However, the impact of mobile learning on students' performance remains equivocal in many other studies. For example, Williams (2009) did not obtain significant results to support the superiority of mobile learning over conventional face-to-face classroom learning. Also, Chu (2014) evaluated the effectiveness of implementing mobile learning in a course for fifth-grade students at an elementary school Taiwan, and found that mobile learning caused a high cognitive load for students and had negative effects on their learning achievements. There were different limitations in the settings of previous studies, leaving research gaps for

further exploration. For instance, in some studies, the participants did not have a strong incentive to use mobile devices throughout a long period of time – their motivation to keep using the mobile devices seemed weak (Vogel et al., 2009). To conduct such studies effectively, students must be placed in a real learning environment in which their learning outcomes are important. The significance of these research studies may often be limited by not having a large enough group of participants practising mobile learning over a sufficiently long period of time, thus not getting results close to real-life practices and with sufficient statistical power. For instance, Park, van Neste-Kenny, Burton, and Kenny (2010) found that the students were not even able to master the use of mobile devices for learning over a five-week period. Therefore, there is a need to assess the effectiveness of mobile learning which is implemented in a real setting for an extended time period among a relatively large group of students (Kenny, van Neste-Kenny, Park, Burton, & Meiers, 2009). The present study, in which mobile learning was situated in a real setting among over 200 students and had been practised for at least one academic semester, represents an effort to fill the gap in previous research.

Learning motivation has not been well elaborated as a concept in models related to mobile learning. For example, in Koole's FRAME model (2009), mobile learning is effective when three aspects – the device, learner and social factors – are all covered, and where motivation is only one of the attributes in the learner aspect without further specification. Similarly, Yau and Joy (2010) proposed a model consisting of five dimensions relevant to students' preferences for mobile learning. The learners' level of motivation is one of the dimensions where possible factors are suggested, such as intrinsic and extrinsic motivations, without further validation.

Meanwhile, there have been extensive studies on learning motivation. It has been viewed as central to engagement in learning (Appleton, Christenson, & Furlong, 2008; Furrer & Skinner, 2003), and consists of a variety of components following two dominant theories in explaining motivation, namely achievement goal theory and expectancy-value theory (Liem, Lau, & Nie, 2008). In the achievement goal theory, students have different purposes or goal orientations for engaging in academic tasks, including (1) performance goal orientation – demonstration of competence relative to others and (2) mastery goal orientation – the development of competence and mastery of tasks (Elliot, 1999; Linnenbrink & Pintrich, 2002). The expectancy-value theory suggests that students' achievement and engagement are influenced by their perceived value of tasks and expectations of success (Pintrich & Schunk, 2002). The perceived value of a task includes three components, namely the (1) intrinsic value – the enjoyment gained from doing the task; (2) attainment value – how important it is to do well in the task; and (3) utility value – how useful the task is in relation to one's future (Berger & Karabenick, 2011). A number of studies in the field used the expectancy-value theory to examine the effectiveness of practices. For example, McLoughlin, Lee, and Chan (2007) utilised the theory to evaluate the effect of educational podcasting on students' motivation and achievement. Phillips and Loch (2012), drawing insights from the theory, examined the usefulness of tablet technology in a finance course and identified the strengths of such technology in enhancing students' engagement in learning. On the other hand, Perry and Steck (2015) used the theory to study the incorporation of mobile technology into the teaching of a secondary-level geometry course, and found that the use of iPads hindered the academic proficiency of students. The expectancy-value theory, offering a multidimensional perspective on the interaction between various factors, is adopted in the present

study to understand the effects of mobile apps on the learning motivation of nursing students in Hong Kong.

Methodology

This project's aim was to identify the effects of mobile learning on students' learning motivation, social interaction and study performance in the context of nursing education. It was conducted in a university in Hong Kong, which has been extending the provision of mobile learning to its nursing courses (Li, Lee, Wong, Yau, and Wong, 2015a, 2015b). For learning purposes, each new nursing student is given a tablet which includes mobile learning apps developed for the nursing courses, with specific features such as a clinical glossary, videos of nursing procedures and real-time quizzes.

This research examines students' learning motivation, social interaction and study performance before and after using mobile learning. With their written consent, two cohorts (2014 and 2015) of Year-2 undergraduate nursing students from a course on Health Assessment were recruited for this study. They include a total of 184 students from the 2014 cohort (male: 19.6%; female: 80.4%) and 225 students from the 2015 cohort (male: 23.6%; female: 76.4%). The course was delivered in a face-to-face learning mode in the classroom environment. Unlike the 2015 cohort, the 2014 cohort had not yet practised mobile learning. Except the mobile learning part, the two cohorts of participants received similar learning experiences in various aspects, such as teaching materials, expected learning hours, teacher assignment and assessment.

Data on the learning motivation were collected through (1) a focus group interview with 20 participants focusing on their experiences after practising mobile learning; and (2) a questionnaire survey after using mobile learning for a semester. The questionnaire was adopted from a validated instrument from a previous study by Wong and Li (2013), which was adapted to the mobile learning context for this study. It consisted of 12 items covering six motivational factors, i.e. performance goal orientation, mastery goal orientation, intrinsic value, attainment value, utility and self-efficacy.

The students' social interaction was also examined through the focus group interview. In addition to the 20 student participants, the teacher of the nursing course also took part in the interview.

They were asked about their experience of social interaction among the students, teachers and mentors with mobile devices within and outside the classroom, as well as in the ward during practicums. The students' study performance was examined using a test of their knowledge, which included 30 questions on the comprehension and application of nursing concepts, which were taught in class and covered in their mobile learning apps. The test was conducted at the end of the course for the 2014 cohort (without practising mobile learning); and at both the start and end of the course for the 2015 cohort (before and after practising mobile learning).

Results

The results are based on both qualitative analysis of the students' responses in the focus group interview, and quantitative analysis of the questionnaire survey data, together with assessment of the students' study performance. The findings address three major areas: (1)

the effect of mobile apps on students' learning motivation; (2) the effect of mobile apps on students' study performance; (3) the effect of mobile learning on students' social interaction; and (4) the relationship between students' learning motivation and study performance.

Students' learning motivation

To understand the students' motivation and/or obstacles to practising mobile learning, a triangulation method was adopted, covering both qualitative analysis of the interview data from the students, and quantitative analysis of survey data about students' motivation in mobile learning.

Focus group interview

The responses of students with experience of mobile learning revealed the benefits from and obstacles in using mobile learning, and improvements which could be made to help in transforming positive attitudes towards mobile learning, if any, to actual practice among the students. Three aspects of the responses from the students are summarised below.

First, the students were generally satisfied with the nursing videos provided in the mobile app for revisiting relevant skills in clinical practicums. They pointed out that the video content helped them to visualise the procedures in real practice and facilitated their retention of the content. The mobile app was seen to be useful throughout the whole learning cycle, from engaging in in-class quizzes to after-class revision. Student 'ES' said:

I will use [the nursing video app] for review purposes, because the app contains a lot of videos and then I can practise the nursing skills using the app.

Second, the mobile app used for clinical assessment in wards promoted interaction between students and mentors. Student 'CH' explained:

The clinical mentor can use the assessment app to monitor our study progress and give feedback immediately. When we are practising wound dressing, for example, the clinical mentor can check which students have passed the assessment on that item through the mobile app, and focus more on helping those students who need more assistance.

Third, institutional support, such as technical assistance and warranty of the devices throughout the students' study period, was highly recommended by the students, as this motivated their use of the mobile devices without worrying about bearing the maintenance cost on their own. The students indicated the technical problems they encountered, such as unstable connection of the device to the server. For example, student 'M' talked about his experiences:

There was a time I have completed a clinical assessment using the app, but the system displayed that I have not completed the assessment items. I needed to call the technical support staff for help. This is important as sometimes there are technical problems with the mobile app we cannot solve.

Questionnaire survey

The students' motivation in mobile learning was also shown in their ratings on 12 questionnaire items on various motivational factors. Table 1 shows the mean ratings of the items (using a seven-point Likert scale ranging from 1 'strongly disagree' to 7 'strongly agree'). Table 2 shows the mean ratings on the six motivational factors.

Table 1. Mean ratings on items about students' motivation in mobile learning.

Rank	Item #	Item content	Mean	SD	n
1	11	It is important for me to have an outstanding academic performance	4.92	1.14	225
2	10	If I am interested in the course, I still want to learn no matter whether the course content is difficult or not	4.86	1.18	225
3	7	I think these mobile apps can be used as supplementary materials for other nursing courses	4.76	1.23	223
4	9	I want to learn as much as possible in this course.	4.73	1.06	224
5	4	I enjoy self-paced learning by using the mobile apps	4.68	1.13	225
6	12	Outperforming my classmates is my learning motivation	4.65	1.16	225
7	5	Using these mobile apps for study can enhance my understanding of health assessment	4.61	1.13	225
8	3	These mobile apps arouse my interest and motivate my self-study	4.59	1.24	225
9	6	My learning experience with these mobile apps adds value to my overall academic performance	4.53	1.07	225
10	2	My level of satisfaction with these mobile apps for health assessment is high	4.47	1.19	224
11	1	I have learnt a lot from these mobile apps	4.40	1.22	224
12	8	I think mobile learning is better than classroom learning	4.09	1.37	225

Table 2. Average ratings on items about students' motivation in mobile learning.

Rank	Items #	Factor	Mean	SD	n
1	9, 10	Mastery goal orientation	4.79	1.01	225
2	11, 12	Performance goal orientation	4.79	1.05	225
3	3, 4	Attainment value	4.63	1.12	223
4	5, 6	Self-efficacy	4.57	1.02	224
5	1, 2	Intrinsic value	4.55	1.06	225
6	7, 8	Utility	4.42	1.11	225

The results showed various extents of effects of the mobile learning practice on the motivational factors. The students gave relatively high ratings for both mastery goal orientation and performance goal orientation. They considered it was important to obtain outstanding academic performance and to outperform others. They also showed strong motivation to master the content of the course and learn as much as possible.

The factors related to the perceived value and expectation of success in mobile learning – intrinsic value, utility and self-efficacy – were rated relatively low. While the students enjoyed the self-paced learning through the mobile apps, the perceived level of self-efficacy and the added value to learning interest and performance were ranked at a medium level. In addition, the students' satisfaction with the mobile apps, the overall good learning outcome from using it, and the relative advantage of mobile learning over classroom learning were rated the lowest.

There is a statistically significant difference between the high and low ratings for motivational factors. For example, comparing the ratings for mastery goal orientation and utility using an independent sample *t*-test, $t(448)=3.70, p < .01$, the result suggests that the students' practice of mobile learning would have varying degree of effect on different aspects of their learning motivation.

Social interaction

The interaction among students and teachers in class was revealed through qualitative analysis of the focus group interview data. Prior to the mobile learning practise, the in-class student–teacher interaction was at a low level. The interviewees stated that the classroom settings did not encourage interaction between students and teachers. Students ‘Ng’ said:

The teachers sometime asked questions about the contents of previous lecture, and usually those students sitting in the first few rows answered those questions. It is somehow not feasible to ask those sitting at the back to shout their answers out. Answering the questions correctly would definitely be commendable, whereas spelling out a wrong answer would be very embarrassing.

The students also shared that they seldom approached their teachers with questions. Instead, they relied heavily on the course materials given to them for learning, together with online resources where their reliability was one of the concerns the students had.

When asked about the perceived effects of mobile learning on social interaction, the participants said that the use of mobile devices enabled more variety of teaching methods which were helpful in promoting student–teacher interaction – yet the in-class interaction needed to be supported by infrastructure, such as reliable network connection. The teacher of the nursing course pointed out that:

We had a mobile polling system to facilitate interactive learning. It however depended on a good network to support the system, as well as a high computing power of the students’ mobile devices in order to have a smooth connection. It sometimes took us much time in handling technological issues, and students in the class were not satisfied with that situation, conceiving that as a waste of time.

The mobile app features an immediate Q and A function, which served as a channel for student–teacher interaction to facilitate students’ participation in class. The mobile app fostered interaction learning (student–student, student–teacher and student–content) and enabled immediate feedback to students. Student ‘Sit’ responded:

Using the mobile app can facilitate the Q and A in class in an anonymous manner. Many students are afraid of being teased by others if they ask stupid questions.

During the clinical practicum, the students needed to have intensive communication with, and be assessed by, their mentors using a mobile app specifically developed for this. The participants stated that sufficient and clear instructions are needed to use the mobile app effectively for the assessment. Students ‘J’ and ‘S’ said:

Some of the assessment items are not clear. I had an experience that my mentor asked me about the meaning of an assessment item [about] which I did not have a clue either. The item was too brief that made [it] hardly comprehensible. (Student ‘J’)

Some students encountered problems with the use of the devices. The shift from paper-based to mobile-based assessment needed time to let both students and mentors become familiar with the app, so that the interaction with mentors can focus more on the assessment. (Student ‘S’)

The mobile app can facilitate interaction between students and teachers, through allowing the teachers to design more interactive class exercises and discussions. Yet, the size of a class also serves as an important factor affecting the interaction. Teacher ‘SL’ mentioned:

There are many students in a lecture. To help them participate and concentrate in class, I allocate some time in a lecture for class exercises using the mobile app. There are much fewer students in a tutorial and I asked them to do the exercises again using the app and under such a setting. I am able to talk with them individually.

Table 3. Comparison of test scores between two learning groups.

Test score*	Conventional learning group (<i>n</i> = 184)		Mobile learning group (<i>n</i> = 202)		<i>t</i>	<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
Comprehension	9.66	2.37	9.57	1.86	−0.44	.66
Application	9.82	2.12	10.46	1.87	3.15	<.01
Total	19.48	3.91	20.03	3.03	1.54	.12

*Full score: 15 for comprehension and application, and 30 in total.

Table 4. Pre-test–post-test analysis of test scores before and after mobile learning.

Test score*	Before mobile learning		After mobile learning		<i>n</i>	<i>t</i>	<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			
Comprehension	8.56	2.01	9.60	1.88	180	5.21	<.01
Application	8.86	2.02	10.46	1.86	180	7.86	<.01
Total	17.33	3.67	20.07	3.05	181	7.75	<.01

*Full score: 15 for comprehension and application, and 30 in total.

Study performance

The study performance of students practising mobile learning was shown by group comparison of the test scores between the students who had not practised mobile learning (2014 cohort) and those who had practised mobile learning (2015 cohort). A pre-test–post-test analysis of the test scores was also conducted for the same group of students (2015 cohort) before and after using mobile learning.

Group Comparison

Table 3 shows the test scores for the conventional learning group and the mobile learning group, and the results of an independent sample *t*-test. Of the three types of scores, only the application score showed a significant group differences – $t(384) = 3.15, p < .01$ – indicating that the mobile learning group outperformed the conventional learning group in terms of the application of nursing concepts.

Pre-test–post-test Analysis

A pre-test–post-test analysis was conducted on the same group of students before and after practising mobile learning. It addresses the limitations of the group comparison between two different cohorts – the differences in test scores might be due to other between-group differences that are relevant to the study performance, for example, different learning ability of two naturally occurring groups, but not the effect of mobile learning.

Table 4 presents the results for a paired sample *t*-test for the test scores. The results show that the students had a significant improvement in all test scores after using mobile learning. The mobile learning practice helped the students to obtain a better learning outcome.

Relationship between students' motivation and study performance

A correlation analysis was carried out between students' motivation ratings and their learning performance in order to determine whether the students' study performance after practising mobile learning was related to students' motivation in mobile learning.

Table 5. Correlations between students' learning motivation and study performance.

	Test score		
	Comprehension	Application	Total
Motivation ratings	<i>r</i> (200)	<i>r</i> (200)	<i>r</i> (200)
Motivation (aggregate)	.05	.08	.08
Mastery goal orientation	.08	.03	.06
Performance goal orientation	.09	.06	.06
Attainment value	-.01	.10	.05
Self-efficacy	.03	.07	.06
Intrinsic value	.04	.11	.10
Utility	-.01	.01	-.00

Table 6. Correlations between students' motivation and study performance improvement.

	Improvement in test score		
	Comprehension	Application	Total
Motivation ratings	<i>r</i> (178)	<i>r</i> (178)	<i>r</i> (179)
Motivation (aggregate)	.140	.08	.13
Mastery goal orientation	.15	-.00	.09
Performance goal orientation	.12	.02	.05
Attainment value	.11	.12	.15
Self-efficacy	.13	.09	.13
Intrinsic value	.12	.11	.15*
Utility	.10	.07	.11

* $p < .05$.

Correlations between learning motivation and study performance

A correlation analysis was conducted between the students' motivation and the test scores for the students after practising mobile learning. Table 5 presents the Pearson correlation coefficients for each pair of variables, including the correlations of each of the three test scores with the aggregate score for motivation rating (taking the mean of the 12 questionnaire items) and the sub-score for each of the six factors. No significant correlations were found, indicating that the test performance may be predicted better by other variables.

Correlations between motivation and improvement in study performance

Another correlation analysis was conducted to test whether the students' motivation was related to the extent of improvement in study performance. This analysis focused on the amount of progress made by each individual participant after practising mobile learning, which may be more specifically related to motivation to practise mobile learning. The improvement of each participant's study performance was measured by subtracting the test scores before mobile learning from the scores after mobile learning. The improvement in test scores was then correlated with the motivation scores.

Table 6 presents the Pearson correlation coefficients for each pair of variables. The results show a significant correlation between improvement in total scores and the sub-score for intrinsic value: $r(179) = .15, p < .05$. Despite the weak positive association indicating that such improvement may be related to other variables, the result suggests that it is important for motivating students in terms of their perceived intrinsic value – the enjoyment gained from practising mobile learning – to gain improvement in study performance.

Discussion

The findings showed the potential effect of mobile apps on students' learning motivation, particularly in terms of complementing conventional classroom learning and making up for its deficiencies. The participants, who were situated in a real learning context, maintained a relatively high motivation for practising mobile learning. This addresses the limitations in previous research where mobile learning was practised in an experimental setting (Kenny et al., 2009; Vogel et al., 2009). The students' feedback highlighted the advantages of mobile learning over conventional face-to-face classroom learning, which provided them with incentives to engage in mobile learning. For example, the students' expectations for multimedia and interactive learning materials were satisfied according to those who had practised mobile learning. Particularly in clinical practicums, the students deemed the learning content useful and significant in evoking their sensory and cognitive curiosity, and contributing to their enjoyment (Liu, Toprac, & Yuen, 2009; Malone & Lepper, 1987). This confirms the effectiveness of the use of mobile technologies in promoting just-in-time and contextual learning which have been emphasised in nursing education (Chan, 2002; Rees et al., 2015). The mobile learning practice was shown to have tackled the technological and policy challenges as stated in Khaddage et al. (2015) regarding the need for apps developed for specific courses and the provision of technical support. The positive students' experience suggests how the effectiveness of the practice enhanced their motivation in learning.

Among the various features of the mobile apps, the students found the supplementary information particularly useful and enjoyed the self-paced learning supported by mobile technologies. Their engagement with mobile learning, according to Pachler et al. (2010), is related to sociocultural conditions – structure, cultural practice and agency. The structure of the nursing education institution, as represented by the intense adoption of mobile apps in the teaching and learning, played an indispensable role in influencing students' acceptance of mobile learning. The growing prevalence of mobile means for communication and sharing of information, which has become an established cultural practice among students, also contributed to the incorporation of mobile technologies into the classroom. Meanwhile, the nursing teachers, who actively mediated the interaction between conventional teaching and assessment approaches and new technologies, helped in promoting the recognition of the value of mobile learning. In line with previous studies (e.g. Denk, 2007), this research shows that mobile learning in the present sociocultural context is useful in supplementing, rather than replacing, conventional learning for enriching the variety of teaching and learning as well as increasing students' motivation.

Both the students and the teacher mentioned that the mobile apps were helpful in facilitating social interaction. In practicums, students and mentors can keep track of the students' learning progress towards desired goals through the performance record feature of the mobile app for clinical assessment, which helped them to communicate constructively in a way conducive to students' mastery of clinical knowledge and skills. In the classroom environment, the mobile app with the polling function facilitated in-class interaction and enabled the use of a broader range of pedagogy.

The effective social interaction may in turn help to foster the students' intrinsic motivation in learning. With the aid of the mobile apps, the students can manage their learning progress more effectively, get helpful feedback from mentors and participate in in-class discussion more comfortably. In the questionnaire survey, the students gave relatively high 'satisfactory'

ratings for the items related to the self-paced learning. Personal control over study can improve students' motivation since it enhances the sense of autonomy and promotes self-determination over meaningful outcomes (Ciampa, 2014; Malone & Lepper, 1987).

The students had a better study performance after practising mobile learning. The findings showed that such improvement was correlated with the students' learning motivation in terms of their intrinsic value in mobile learning, that enrich the present understanding of the impact of mobile learning on students' motivation. Instead of enhancing motivation in general (Gregory & Crompton, 2016; Vogel et al., 2009), mobile learning may improve one or more specific aspects of motivation. In this research, it was found that the intrinsic motivation of students had been improved by the mobile's apps through constant evaluation of one's progress, enhanced curiosity in learning and strengthened control over study. In this regard, the design of mobile learning activities could have more specific objectives in relation to motivating students to learn. Also, as intrinsic value was found to be the only motivational factor associated with improvement in performance, there are potentially other factors leading to the improvement. This suggests that there should be future work on exploring the effect of having higher student motivation in mobile learning and its influence on students' learning outcomes.

This study also points out possible linkage between relevant frameworks of mobile learning and motivation theories. While motivation has yet to be well elaborated in mobile learning models such as Koole's FRAME model (2009) and Yau and Joy's five dimensional model (2010), the findings in this study showed that motivational factors as identified in the achievement goal theory and expectancy-value theory (Elliot, 1999; Liem et al., 2008; Linnenbrink & Pintrich, 2002) can serve to categorise students' motivation in mobile learning and thus potentially extend the relevant models or frameworks in this aspect.

It should be noted that the participants of this study, who were mostly female, may have effects on the study's results. Gender difference has been identified as a potential factor influencing learners' self-efficacy in using digital devices (Bao et al., 2013), and moderating the social influence on usage intention of mobile learning (Wang et al., 2009). Different from the related studies, mobile learning was a compulsory and integral part of the course for the participants of this study. The effect of such a context on the possible gender different in mobile learning is worth further study.

Conclusion

The study presented in this paper highlights the potential effects of mobile apps on learning motivation, social interaction and study performance in the context of nursing education. It demonstrates the importance of conducting mobile learning research in a real setting for an extended period of time and the significance of implementing mobile learning practice with the potential technological and policy challenges taken into consideration. These factors affect students' experience in and incentives for engaging in mobile learning activities. The results of this study reveal the possible effect of mobile learning on different motivational factors, and show that not each motivational factor may be related to study performance. Hence, the flexibility and interactivity of mobile learning are best to be leveraged to supplement, instead of replace, conventional learning at the current stage.

These findings contribute to advancing mobile learning research in refining aspects of both theory and implementation. They suggest possible links between existing frameworks

of mobile learning and theories of motivation, and their influence on study performance. Future work in this regard will lead to a better theoretical foundation for mobile learning and more effective measures for motivating student engagement.

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